Numerical Methods - I

The expression $f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$ where a's are constants $(a_0 \neq 0)$ and n is a positive integer is called a polynomial of degree "n" in x. The Polynomial $f(x)=0$ is called an algebraic equation of degree "n".

If $f(x)$ contains some other functions such as trignometric, logarithmic, exponential etc then $f(x)=0$ is called an transcendental equation.

Ex:- $3x - \cos x - 1 = 0$, $e^x - 3x = 0$, $\sin x = \frac{x}{2}$

The value of x which satisfies $f(x)=0$ is called a root.

Geometrically the root of an equation is the value of x where the graph of $f(x)$ crosses the x-axis.

The process of finding the roots of an equation is known as Solution of the equation.

Simple root:- A number "α" is a simple root of $f(x)=0$, if $f(\alpha)=0$ and $f'(\alpha) \neq 0$. Then we can write $f(x) = (x-\alpha)g(x)$

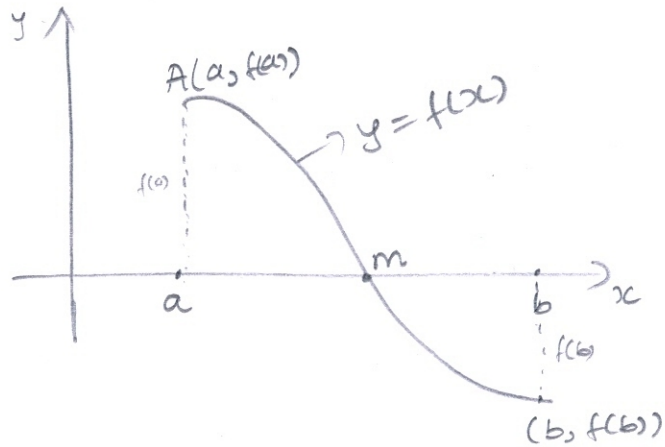
Ex:- $x=1$ is a simple root of $f(x) = x^3 + x - 2$

Since $f(1)=0$ and $f'(1) = 4 \neq 0$.

Multiple root:- A number α is a multiple root, of multiplicity "m" of $f(x)=0$, if $f(\alpha)=0, f'(\alpha)=0, \ldots f^{(m-1)}(\alpha)=0$ and $f^{(m)}(\alpha) \neq 0$

Ex: $x=2$ is a Multiple root of Multiplicity 2 of $f(x) = x^3 - 3x^2 + 4$ since $f(2)=0$, $f'(2)=0$ and $f''(2) = 6 \neq 0$.

Intermediate Value property OR Location of roots Property.



If $f(x)$ is a real valued continuous function, an important property of such function is that $f(x)$ has opposite signs for two values of a and b then the equation $f(x) = 0$ has atleast one root between a and b.

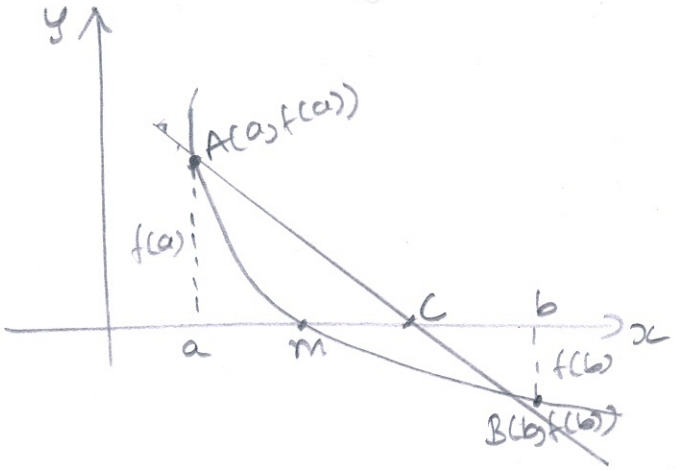
Geometrically it means that if $f(a)$ and $f(b)$ are of opposite signs then the curve $y = f(x)$ cuts the x-axis atleast at one point m that lies between a and b.

The point "m" is a real root of the equation $f(x) = 0$.

Regula-Falsi method or False position method.

Suppose an equation $f(x) = 0$ is known to have only one root m between a and b. this happens when $f(a)$ and $f(b)$ are of opposite signs and the curve $y = f(x)$ crosses the x-axis only once at the point $x = m$ lying between the points $x = a$ and $x = b$.

Consider the points $A(a, f(a))$ and $B(b, f(b))$ on the curve $y = f(x)$.



The equation of the chord AB is

$$\frac{y-f(a)}{x-a} = \frac{f(b)-f(a)}{b-a} \quad \rightarrow (1)$$

At the point C, where the chord AB crosses the x-axis, we have $y=0$ and eqn (1) becomes

$$\frac{-f(a)}{x-a} = \frac{b-a}{f(b)-f(a)} \quad \frac{f(b)-f(a)}{b-a}$$

$$\frac{x-a}{f(a)} = \frac{a-b}{f(b)-f(a)} \qquad \text{[reciprocating]}$$

$$x-a = \frac{af(a)-bf(a)}{f(b)-f(a)}$$

$$x = \frac{af(b)-af(a)+af(a)-bf(a)}{f(b)-f(a)}$$

$$x = \frac{af(b)-bf(a)}{f(b)-f(a)}$$

This gives the x-coordinate of the point C. If the interval $[a,b]$ is sufficiently small, the x-coordinate of the point C is sufficiently close to the point $x=m$.

Formula for Regula-Falsi method

$$x = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

(1) The equation $x^4 - x - 10 = 0$ has one root between 1.8 and 2. Find the root correct to 3 decimal places by the method of false position.

Soln:- $f(x) = x^4 - x - 10$

$a = 1.8 \qquad b = 2$

$f(a) = -1.3024 \;,\quad f(b) = 4$

Thus $f(a)$ and $f(b)$ are of opposite signs.

The first approximation is given by

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

$$= \frac{(1.8)(4) - 2(-1.3024)}{4 - (-1.3024)}$$

$$= 1.8491$$

$$f(x_1) = -0.1583$$

$[\because f(1.849) = -0.1583$ -ve, $f(2) = 4$ +ve$]$

Thus the root lies between 1.8491 and 2.

Now $a = 1.8491$ and $b = 2$

$f(a) = -0.1583 \qquad f(b) = f(2) = 4$

∴ Second approximation is

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

$$x_2 = \frac{(1.8491)(4) - 2(-0.1583)}{4 - (-0.1583)}$$

$$x_2 = 1.8548$$

$$f(x_2) = -0.0192$$

Thus the root lies between 1.8548 and 2

Now $a = 1.8548$ and $b = 2$

$f(a) = f(1.8548) = -0.0192$, $f(b) = f(2) = 4$

∴ third approximation is

$$x_3 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

$$= \frac{1.8548(4) - 2(-0.0192)}{4 - (-0.0192)}$$

$$= 1.8554$$

[$f(1.8554) = -0.0045$ Almost zero]

Hence the approximate root after the 3rd approximation is 1.8554.

(2) Find the root of the equation $2x - \log_{10} x = 7$ which lies between 3.5 and 4 by regula-falsi method.

Soln: Let $f(x) = 2x - \log_{10} x - 7$

Here $a = 3.5$ and $b = 4$

$f(a) = -0.5440$, $f(b) = 0.3979$

In Scientific calculator
log – $\log_{10} x$
ln → $\log_e x$

We note that $f(a)$ and $f(b)$ are of opposite signs and hence root lies between 3.5 and 4

First approximation

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{3.5(0.3979) - 4(-0.5440)}{0.3979 + 0.5440}$$

$$= 3.7887$$

$$f(x_1) = -0.0010$$

Second approximation

we take $a = 3.7887$ $b = 4$

$f(a) = -0.0010$ $f(b) = 0.3979$

$$\therefore x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{3.7887(0.3979) - 4(-0.0010)}{0.3979 - (-0.0010)}$$

$$= 3.7891$$

Since $f(x_2) = f(3.7891) = -0.0005$ Almost zero.

we take 3.7891 as the approximated root for the given equation.

(3) Using Regula-Falsi method, find a real root of the equation $\cos x = 3x - 1$ that lies between 0.5 and 1.

Soln:- Let $f(x) = \cos x - 3x + 1$ "Set calculator mode to radians"

Here $a = 0.5$ and $b = 1$

$f(a) = f(0.5) = 0.3775$, $f(b) = f(1) = -1.4596$

First approximation

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0.5(-1.4596) - 1(0.3775)}{-1.4596 - 0.3775}$$

$x_1 = 0.6027$

$\therefore f(x_1) = 0.0157$

For the second approximation, we take

$a = 0.6027 \qquad b = 1$

$f(a) = 0.0157 \qquad f(b) = -1.4596$

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0.6027(-1.4596) - 1(0.0157)}{-1.4596 - 0.0157}$$

$$= 0.6069$$

Now $f(x_2) = f(0.6069) = 0.0007$

Hence the approximate root is 0.6069

④ Using Regula-Falsi method, determine a positive root of $e^x = 3x + \sin x$ correct to 4 decimal places.

Soln: Let $f(x) = e^x - 3x - \sin x$

Taking the trials, $f(0) = 1$, $f(1) = -1.1231$

So, a root of $f(x)$ lies between 0 and 1.

Now $a = 0$, $b = 1$, $f(a) = 1$, $f(b) = -1.1231$

First approximation,

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0(-1.231) - 1(1)}{-1.1231 - 1}$$

$$= 0.4710$$

$f(x_1) = -0.2651$

Now the root lies between 0 and 0.4710 since

$f(0) = 1$ and $f(0.4710) = -0.2651$

i.e they are of opposite signs.

Second approximation

$a = 0$, $b = 0.4710$, $f(a) = 1$, $f(b) = -0.2651$

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0(-0.2651) - 0.4710(1)}{-0.2651 - 1}$$

$$= 0.3723$$

Thus $f(x_2) = -0.0295$

Now the root lies between 0 and 0.3723 since

$f(0)$ and $f(0.3723)$ are of opposite signs.

Third approximation :- $a = 0$, $b = 0.3723$, $f(a) = 1$

$f(b) = -0.0295$)

$$x_3 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

$$= \frac{0(-0.0295) - 0.3723(1)}{-0.0295 - 1}$$

$$= 0.3616$$

$$f(x_3) = f(0.3616) = -0.0029$$

Hence the approximate root of the given equation is 0.3616